LESSON PLAN

A/C MATERIAL (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	15
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Panel bucking, estimation of compressive loads for bucking a panel on simply supported structures
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Panel bucking, estimation of compressive loads for bucking a panel on simply supported structures. Rigidity due to end conditions on panel, study of heads, to bend panel with regards to variation in end condition shear buckling of plates
5. Specific Objectives	15.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of theoretical buckling loads and end condition rigidity factors using IB diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary failure mode for thin skin panels under compression?
(b) Define the term 'buckling' in the context of aircraft structures.
(c) List three common structural elements found in aircraft fuselage panels.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of panel bucking • Simply supported structure characteristics • Estimation of compressive loads • Rigidity due to end conditions • Effect of shear buckling on plates	L & Ds	PPT, IB	15 min
(b)	• Study of heads to bend panel • Variation in end condition rigidity • Calculation of critical buckling stress • Influence of boundary conditions • Comparison of theoretical vs actual loads	L & Ds	PPT, PDF	20 min
(c)	• Impact of shear buckling on plate stability • Role of stiffeners in preventing buckling • Analysis of load distribution in panels • Factors affecting compressive load estimation • Summary of end condition effects	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how end conditions affect the rigidity of a panel under compressive load.
(b) Describe the relationship between shear buckling and the overall stability of a plate.
(c) Identify the key factors that must be considered when estimating compressive loads for simply supported structures.

12. Home Assignment and Follow Up	Prepare a brief note on how the variation in end conditions influences the critical buckling load, referencing concepts covered in LP 16 for damage assessment.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Assessment of damage and estimation of loads, preparation of repair scheme. Material specification, special tools required for carrying out repairs	Refer to the master syllabus reference list	LP 16

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min